

SO(10) anomaly cancellation via the spinor-trace method: five Standard-Model perturbative anomaly coefficients + one mixed gravitational coefficient are symbolically exact zero per generation (auxiliary anomaly-cancellation note)

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Within the SO(10) grand-unification framework as employed by the Topological Energy Condensate Theory (TECT) Pillar 4 candidate embedding (Math157), we audit the per-generation cancellation of all Standard-Model perturbative gauge anomalies and of the mixed gravitational $U(1)_Y$ anomaly. The five perturbative coefficients

$$\text{Tr}(U(1)_Y^3), \quad \text{Tr}(SU(2)_L^3), \quad \text{Tr}(SU(3)_c^3), \quad \text{Tr}(U(1)_Y \cdot SU(2)_L^2), \quad \text{Tr}(U(1)_Y \cdot SU(3)_c^2)$$

and the one gravitational mixed coefficient $\text{Tr}(U(1)_Y)$ vanish *symbolically exactly* per generation when summed over the chiral content of the SO(10) **16** spinor reduced under $SU(5) \times U(1)_\chi \rightarrow SU(3)_c \times SU(2)_L \times U(1)_Y$. The SU(5) sub-content $\bar{\mathbf{5}} \oplus \mathbf{10}$ already saturates all six perturbative + gravitational SM anomalies; the SO(10) singlet **1**(5) (the right-handed neutrino, hereafter RHN) carries hypercharge $Y = 0$ and therefore contributes *zero* to any $U(1)_Y$ trace. The RHN's genuine consistency role is (i) closure of the Witten SU(2) global anomaly (non-perturbative mod-2, *not* part of the perturbative six-coefficient list); and (ii) when $U(1)_{B-L}$ is gauged in the SO(10) extension, saturation of the mixed gravitational $U(1)_{B-L}$ anomaly. The result is conditional on (a) the SO(10) embedding of TECT Pillar 4 (Math157) and (b) the standard $SU(5) \rightarrow \text{SM}$ hypercharge assignment via $Y = T_{3R} + \frac{1}{2}(B-L)$. This paper is an independent auxiliary anomaly-cancellation note and is *not* coupled to the unrelated three-generation question (Paper 07). Math319 V1 verification target adopts these six cancellations as its symbolic-anomaly check.

I. INTRODUCTION

Anomaly cancellation is a cornerstone of consistent quantum field theory. For a four-dimensional chiral gauge theory with gauge group $G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y$, perturbative consistency requires the vanishing of five chiral triangle anomaly coefficients,

$$\text{Tr}(U(1)_Y^3), \text{Tr}(SU(2)_L^3), \text{Tr}(SU(3)_c^3), \text{Tr}(U(1)_Y \cdot SU(2)_L^2), \text{Tr}(U(1)_Y \cdot SU(3)_c^2), \quad (1)$$

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together with the vanishing of the mixed gravitational anomaly

$$\text{Tr}(\text{U}(1)_Y), \quad (2)$$

which is the coefficient of the graviton–graviton– $\text{U}(1)_Y$ triangle. The traces in (1)–(2) are sums over all left-handed (or, equivalently, all L -and- R - conjugate) chiral fermions of the theory.

Two consistency conditions of the SM are not entries in (1)–(2):

- the *Witten* $\text{SU}(2)_L$ *global anomaly* [1], which is a non-perturbative $\mathbb{Z}_2$ statement requiring an even number of chiral $\text{SU}(2)_L$ doublets;
- the gauged $\text{U}(1)_{B-L}$ mixed gravitational anomaly $\text{Tr}(B-L)$, which appears *only* when $B-L$ is promoted to a gauge symmetry (as in the $\text{SO}(10)$ extension of the SM).

Both are independent of (1)–(2); their treatment in TECT is recorded below as a separate audit (§IV).

The $\text{SO}(10)$ grand-unification group naturally houses the SM gauge structure. The spinor representation **16** of $\text{SO}(10)$ contains one complete SM family plus one electroweak singlet (the RHN). Math157 exploits this to derive all SM gauge anomaly coefficients from the group-theoretic trace formulas of $\text{SO}(10)$.

II. SETTING AND MAIN RESULT

The $\text{SO}(10)$ spinor **16** decomposes under $\text{SU}(5) \times \text{U}(1)_\chi$ as

$$\mathbf{16} = \mathbf{10}(1) \oplus \bar{\mathbf{5}}(-3) \oplus \mathbf{1}(5), \quad (3)$$

and further to the SM gauge group as

$$\mathbf{10}(1) \rightarrow \underbrace{(\mathbf{3}, \mathbf{2})_{1/6}}_{Q_L} \oplus \underbrace{(\bar{\mathbf{3}}, \mathbf{1})_{-2/3}}_{u_R^c} \oplus \underbrace{(\mathbf{1}, \mathbf{1})_1}_{e_R^c}, \quad (4)$$

$$\bar{\mathbf{5}}(-3) \rightarrow \underbrace{(\mathbf{1}, \mathbf{2})_{-1/2}}_{L_L} \oplus \underbrace{(\bar{\mathbf{3}}, \mathbf{1})_{1/3}}_{d_R^c}, \quad (5)$$

$$\mathbf{1}(5) \rightarrow \underbrace{(\mathbf{1}, \mathbf{1})_0}_{\nu_R^c} \quad (\text{the RHN; SM hypercharge } Y = 0). \quad (6)$$

The hypercharge assignment in (4)–(6) is the standard $\text{SU}(5)$ -derived $Y = T_{3R} + \frac{1}{2}(B-L)$.

Theorem 1 ($\text{SO}(10)$ anomaly cancellation via the spinor-trace method). *Under hypotheses*

(**H1**) the $SO(10)$ embedding (Math157) expressing the SM gauge group and one generation of SM matter as the chiral content of the $SO(10)$ **16** spinor as in (3)–(6);

(**H2**) the standard $SU(5)$ -derived hypercharge assignment $Y = T_{3R} + \frac{1}{2}(B - L)$ on each irreducible component (Math157-AddD),

the five perturbative SM gauge anomaly coefficients of (1) and the one mixed gravitational coefficient (2) are symbolically exact zero per generation:

$$\boxed{\begin{aligned} \text{Tr}(U(1)_Y^3) &= 0, & \text{Tr}(SU(2)_L^3) &= 0, \\ \text{Tr}(SU(3)_c^3) &= 0, & \text{Tr}(U(1)_Y \cdot SU(2)_L^2) &= 0, \\ \text{Tr}(U(1)_Y \cdot SU(3)_c^2) &= 0, & \text{Tr}(U(1)_Y) &= 0, \end{aligned}} \quad (7)$$

where the trace is over the chiral content of one **16** of $SO(10)$ (i.e. one SM generation + one RHN). The RHN **1**(5) contributes zero to every line of (7) because its SM hypercharge is $Y = 0$.

The result is **T6 proved conditional** on (H1), (H2), and the textbook-standard $SU(5)$ representation theory employed in (4)–(5).

Remark 2 (The five-coefficient sub-saturation by $\bar{\mathbf{5}} \oplus \mathbf{10}$). The five perturbative coefficients of (1) are *already* saturated by the $SU(5)$ sub-content $\bar{\mathbf{5}} \oplus \mathbf{10}$ alone: this is a classical result of Georgi–Glashow $SU(5)$ GUT phenomenology [2]. The RHN **1**(5) is unnecessary for (1). Its role in (7) is via $\text{Tr}(U(1)_Y)$; but since the RHN has $Y = 0$, it contributes zero there as well, and the gravitational mixed anomaly is also saturated by $\bar{\mathbf{5}} \oplus \mathbf{10}$ alone.

Remark 3 (Multi-generation extension). For N_g generations the same theorem holds with all coefficients in (7) multiplied by N_g . Cancellation is therefore N_g -independent. Anomaly cancellation *does not* fix the value of N_g . (See §IV on the separate Witten $SU(2)$ global anomaly, which constrains $N_g \bmod 2$ only via the total chiral $SU(2)_L$ doublet count.)

III. PROOF: TERM-BY-TERM SPINOR TRACE

a. Step 1 ($SO(10)$ spinor decomposition). The $SO(10)$ Clifford algebra and the **16** spinor representation are computed via the standard 32×32 Dirac gamma-matrix construction of [3]; the $SU(5) \times U(1)_\chi$ branching (3) is read off from the $SO(10)$ Dynkin diagram via the extended-Dynkin folding [4]. The further reduction to $SU(3)_c \times SU(2)_L \times U(1)_Y$ in (4)–(5) follows the standard $SU(5) \rightarrow \text{SM}$ branching (Math157, [5]).

b. Step 2 (SM hypercharge assignment). Each chiral component receives its SM hypercharge via $Y = T_{3R} + \frac{1}{2}(B - L)$, where T_{3R} is the third component of right-handed isospin (read from the $\text{SO}(10)$ $\text{SU}(2)_R$ sub-algebra) and $(B - L)$ is the baryon-minus-lepton charge (Math157-AddD, [6]).

c. Step 3 (trace formulas). For each anomaly coefficient, the trace is computed by summing the appropriate Dynkin index and charge over all left-handed chiral fermions in the spinor:

$$\text{Tr}(G^a G^b G^c) = \sum_{f \in \mathbf{16}} d_{R_f}^{abc} n_f, \quad (8)$$

where $d_{R_f}^{abc}$ is the symmetric anomaly coefficient of the representation R_f of the chiral fermion f under generators G^a, G^b, G^c , and $n_f = 1$ for left-handed fermions, $n_f = -1$ for right-handed (here we use the L -conjugate convention so all $n_f = 1$).

d. Step 4 (term-by-term cancellation). The five perturbative coefficients of (1) evaluate on the chiral content of $\bar{\mathbf{5}} \oplus \mathbf{10}$ to:

$$\text{Tr}(\text{U}(1)_Y^3) = 3 \cdot 2 \cdot \left(\frac{1}{6}\right)^3 + 3 \cdot \left(-\frac{2}{3}\right)^3 + (1)^3 + 2 \cdot \left(-\frac{1}{2}\right)^3 + 3 \cdot \left(\frac{1}{3}\right)^3 = 0, \quad (9)$$

$$\text{Tr}(\text{SU}(2)_L^3) = 0 \quad (\text{symmetric anomaly trivially vanishes for } \text{SU}(2)); \quad (10)$$

Witten global anomaly is a separate mod-2 statement),

$$\text{Tr}(\text{SU}(3)_c^3) = 2 \cdot 1 + (-1) + (-1) = 0 \quad (\text{net triangle vanishes}), \quad (11)$$

$$\text{Tr}(\text{U}(1)_Y \cdot \text{SU}(2)_L^2) = 3 \cdot \frac{1}{6} \cdot \frac{1}{2} + \left(-\frac{1}{2}\right) \cdot \frac{1}{2} = 0, \quad (12)$$

$$\text{Tr}(\text{U}(1)_Y \cdot \text{SU}(3)_c^2) = 2 \cdot \frac{1}{6} \cdot \frac{1}{2} + \left(-\frac{2}{3}\right) \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} = 0, \quad (13)$$

where the $\text{SU}(2)$ (resp. $\text{SU}(3)$) Dynkin index $\frac{1}{2}$ for the fundamental, and multiplicities follow from (4)–(5). The mixed gravitational coefficient evaluates to

$$\text{Tr}(\text{U}(1)_Y) = 3 \cdot 2 \cdot \frac{1}{6} + 3 \cdot \left(-\frac{2}{3}\right) + 1 + 2 \cdot \left(-\frac{1}{2}\right) + 3 \cdot \frac{1}{3} = 0, \quad (14)$$

again from $\bar{\mathbf{5}} \oplus \mathbf{10}$ alone. The RHN contributes zero everywhere (its $\text{SU}(2)_L$, $\text{SU}(3)_c$, and Y charges are all zero). $\square$

IV. THE RHN'S GENUINE CONSISTENCY ROLE

The earlier draft of this paper (Rev 3) asserted that the RHN cancels the $\text{U}(1)_Y$ anomaly of the $\bar{\mathbf{5}}(-3)$ representation. As recorded in §III and Remark 2, this statement is incorrect: the RHN has SM hypercharge $Y = 0$ and contributes zero to every $\text{U}(1)_Y$ trace, while $\bar{\mathbf{5}} \oplus \mathbf{10}$ already saturates all six SM perturbative + gravitational anomalies of (7).

The RHN does play a real role in three independent consistency audits of the $\text{SO}(10)$ extension:

1. **Witten $SU(2)_L$ global anomaly** [1]. This is a non-perturbative $\mathbb{Z}_2$ statement: a four-dimensional theory with an odd number of chiral $SU(2)_L$ doublets is inconsistent at the path-integral level. The SM has, per generation, the doublets Q_L (with multiplicity 3 from colour) and L_L (multiplicity 1), giving a total of 4 chiral doublets per generation, which is even; the global anomaly already cancels *without* the RHN. The RHN, being an $SU(2)$ singlet, contributes zero doublets and therefore plays no role in the Witten cancellation either. Its consistency role here is *neutral* (it does not break the cancellation; equivalently, $SU(2)$ anomaly cancellation does not require it).
2. **Mixed gravitational $U(1)_{B-L}$ anomaly**. When $B - L$ is gauged in the $SO(10)$ extension (the $U(1)_{B-L}$ subgroup of the $SO(10)$ Pati–Salam $SU(4)_C$), the trace $\text{Tr}(B - L)$ is a new mixed gravitational anomaly. The RHN has $(B - L) = +1$ (lepton number of opposite sign because it is the L -conjugate of ν_R); it is required to make $\text{Tr}(B - L) = 0$ per generation. This is the RHN’s strongest classical role.
3. **$SO(10)$ algebra anomaly freedom**. The Lie algebra $\mathfrak{so}(10)$ is anomaly-free at the simple-algebra level ([4], Sec. 4); its representations admit the spinor **16** as the smallest faithful spinor that contains one full SM generation. The RHN appears as the singlet component of **16**; its presence is the consequence, not the cause, of completing the spinor.

None of (1)–(3) is identified with the perturbative-gauge cancellation (1) or with the gravitational $\text{Tr}(Y)$ cancellation (14). The corrected statement is therefore: *the RHN is required for the $B - L$ -gauged extension and for completion of the **16** spinor; it is not required for SM perturbative or gravitational $U(1)_Y$ anomaly cancellation.*

V. NUMERICAL VERIFICATION

The six trace identities (7) were cross-checked by symbolic algebra over the $SU(5)$ representation content (4)–(5); each line of (7) returns the identically-zero rational number, not a finite-precision estimate. A subsequent numerical evaluation in machine arithmetic returns no residual at machine precision; this is a redundant cross-check and *not* a numerical estimate.

Remark 4 (Symbolic versus numerical exactness). The earlier draft phrasing “numerically to six-digit precision” was imprecise. The cancellation (7) is a symbolic identity in the rational numbers; the appropriate phrasing is “symbolically exact zero, numerically cross-checked”. The Math319 V1 verification target adopts the symbolic form.

VI. DISCUSSION AND OUTLOOK

The anomaly-cancellation result (7) is a non-trivial consistency check of the $\text{SO}(10)$ embedding derived from the BCC condensate (Math157, Math229). It is independent of the three-generation question (Paper 07): cancellation holds per generation, for any number N_g of generations. Anomaly cancellation therefore does *not* fix the SM generation count; that question remains the canonical Pillar 7 **T1 OPEN** problem.

The RHN's classical role is correctly identified, post-correction, as $B - L$ anomaly closure and $\text{SO}(10)$ spinor completion, not as $\text{U}(1)_Y$ saturation (which is entirely absorbed by the $\bar{\mathbf{5}} \oplus \mathbf{10}$ sub-content of the $\text{SU}(5)$ GUT). This paper is the auxiliary anomaly-cancellation note for the TECT Pillar 4 $\text{SO}(10)$ embedding programme; it is decoupled from Paper 07 (three-generation forcing), which carries an independent audit-flag and rewrite obligation.

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 - [6] TECT internal note Math157-AddD, “RHN identification and $B - L$ assignment in the $\text{SO}(10)$ embedding,” TECT canonical archive.

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